
*Government Pension Beneficiary
and Payment Analysis Using
Power BI*

Project Overview

This project involves cleaning, transforming, and analysing raw data using Power BI and creating an interactive dashboard to derive meaningful business insights.

This project analyzes government pension beneficiary and payment data using **Power BI**. It provides interactive insights into beneficiary distribution across states, districts, residence types, and pension schemes. The dashboard evaluates pension payment disbursement through different payment methods and highlights regional trends. Using KPIs and interactive visualizations, it supports effective monitoring of pension scheme performance and data-driven decision-making.

Data Sources

Source Link: https://ndap.niti.gov.in/dataset/6710?filter_id=5546

Domain: Government Welfare / Public Administration / Social Security

This project falls under the **Government Welfare** domain, focusing on the analysis of pension schemes implemented under social security programs.

Objectives

- Analyze the **total number of pension beneficiaries** across different pension schemes.
- Compare **beneficiary distribution** between Rural and Urban areas.
- Evaluate **state-wise and district-wise** beneficiary coverage.

- Analyze **pension payment disbursement** through Bank, Post Office, Money Order, and Cash.
- Identify the **most utilized pension schemes** based on beneficiary count.
- Provide insights to support **government policy evaluation and operational planning**.

Attribute Details

Attribute	Description
Country	Name of the country where the pension scheme is implemented.
State	State in which the pension beneficiaries are located.
District	District within the respective state.
Year	Financial year of the pension data.
Residence Type	Indicates whether beneficiaries belong to Rural or Urban areas.
Scheme Code	Code representing the government pension scheme (e.g., IGNOAPS, IGNDPS, IGNWPS).
Total Number of Beneficiaries	Total number of beneficiaries enrolled under the specified pension scheme.
Pension Payment Through Bank	Number of pension payments disbursed through banks.
Pension Payment Through Post Office	Number of pension payments disbursed through post offices.
Pension Payment Through Money Order Payment System	Number of pension payments disbursed through the money order system.
Pension Payment Through Cash	Number of pension payments disbursed through cash.

Tools & Technologies

Microsoft Power BI Desktop	Data visualization and interactive dashboard development
Power Query	Data cleaning, transformation, and preparation
DAX (Data Analysis Expressions)	Creating calculated columns, measures, and KPIs
Data Modeling	Building relationships and optimizing the data model

Data Pre-Processing (Power Query)

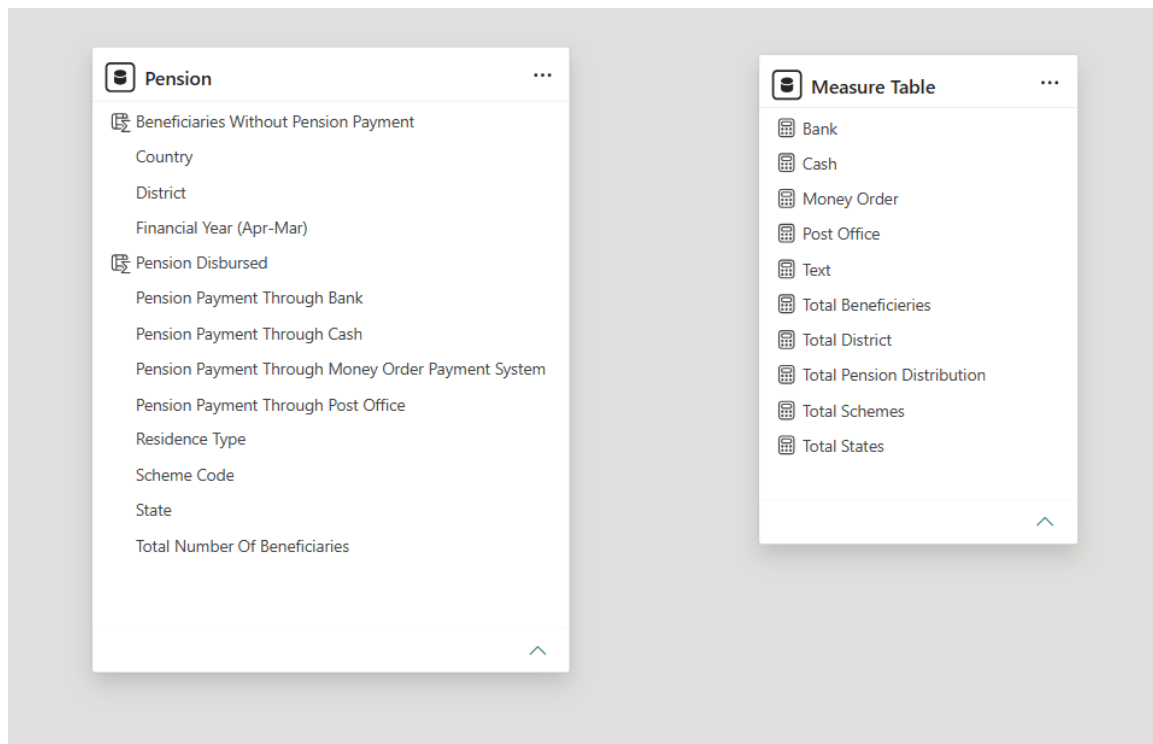
- The dataset was imported into **Power BI** and cleaned using **Power Query** to ensure data consistency.
- Data types were corrected for all columns to enable accurate analysis.
- The financial year information was extracted into a separate **Year** column for trend analysis.
- New calculated columns **Pension Disbursed** and **Beneficiaries Without Payment** were created to support payment analysis. Column names were standardized for better readability and reporting.
- The transformed dataset was then loaded into the Power BI data model for visualization and dashboard development.

Data Modelling and DAX

The dataset was imported into Power BI and organized into a single data model for analysis. The model was structured to support efficient reporting across **State, District, Year, Residence Type, and Scheme Code**.

DAX measures were created to calculate **Total Beneficiaries**, **Pension Disbursed**, and **Beneficiaries Without Payment**.

The data model was optimized for accurate calculations and interactive filtering. This enabled efficient dashboard performance and meaningful business insights.

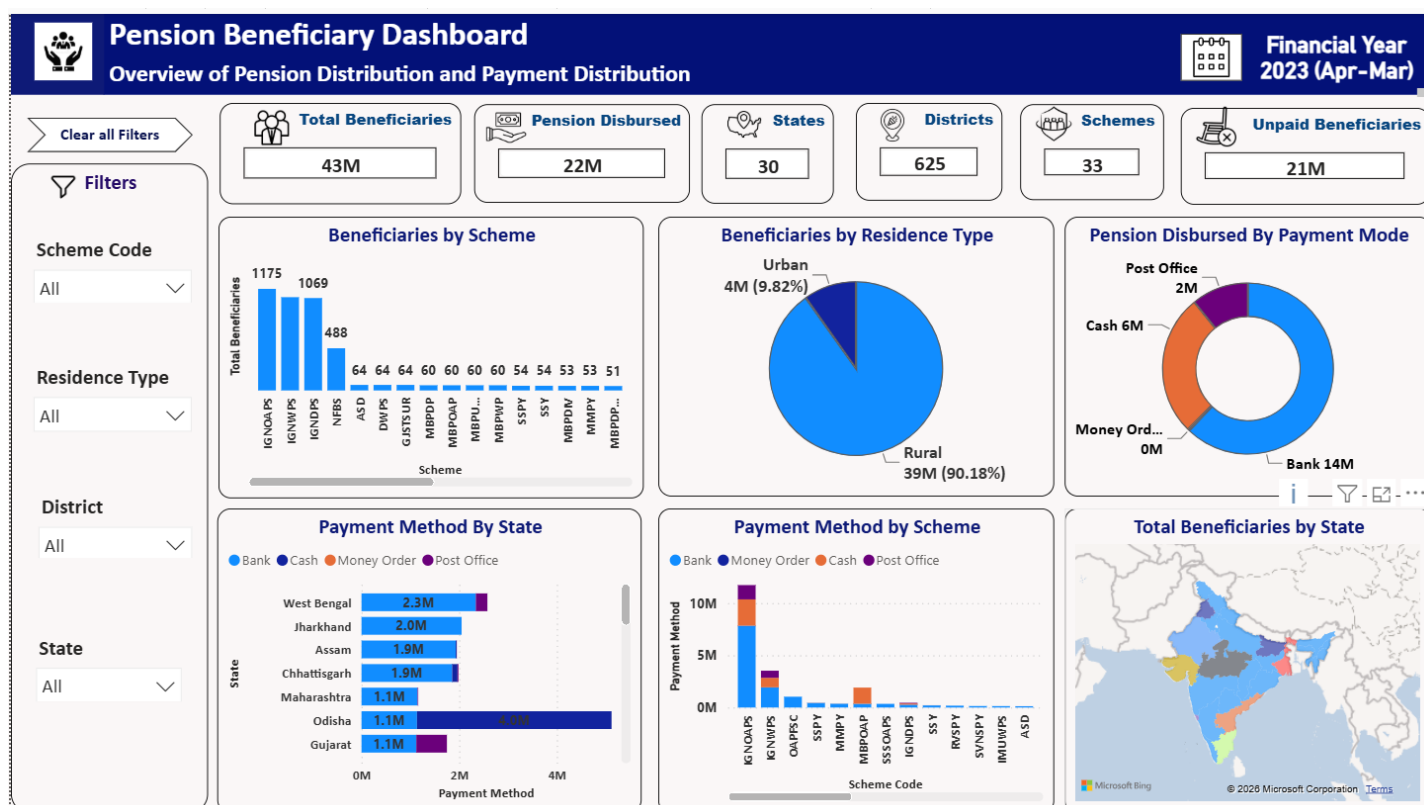


Analysis and Visualizations

The Power BI dashboard was designed using interactive visualizations, including KPI cards, bar charts, Pie charts, maps, and slicers to analyze pension beneficiary data. It provides insights into beneficiary distribution across states, districts, residence types, and pension schemes.

The dashboard also analyzes pension payment disbursement through Bank, Post Office, Money Order, and Cash. Interactive filters and drill-down features enable detailed exploration of the data. The visualizations help identify

beneficiary trends, compare scheme performance, and support data-driven decision-making.



Descriptive Analysis — What Happened?

The dashboard provides an overall view of **43M total beneficiaries** across **30 states**, **625 districts**, and **33 pension schemes**. Out of the dashboard's measures, **22M pension disbursements** are recorded, while the **Unpaid Beneficiaries KPI** shows **21M**.

The beneficiary distribution is highly concentrated in rural areas, with **39M (90.18%)** beneficiaries compared with **4M (9.82%)** in urban areas. Among schemes, **IGNOAPS** leads with **1,175 beneficiary records**, followed by **IGNWPS (1,079)** and **IGNDPS (1,069)**.

The payment-mode analysis shows **Bank** as the leading disbursement channel at **14M**, followed by **Cash** at **6M**, **Post Office** at **2M**, and **Money Order** at **62K**.

Key Numerical Insights

- **43M** total beneficiaries
- **22M** pension disbursed
- **21M** shown as unpaid beneficiaries in the KPI
- **90.18%** of beneficiaries are from rural areas
- **9.82%** are from urban areas
- **14M** through Bank
- **6M** through Cash
- **2M** through Post Office
- **62K** through Money Order
- **625 districts** and **33 schemes** covered

2. Diagnostic Analysis — Why Did These Patterns Occur?

The data indicates a strong rural focus in pension coverage. With **90.18% of beneficiaries residing in rural areas**, pension delivery systems need to be especially accessible in rural locations.

There is also a significant variation in payment methods between states. For example, **Odisha** is particularly notable: although it has the highest beneficiary record count at **748**, its payment distribution shows approximately **4.0M through Cash** compared with **1.1M through Bank**. This is very different from states such as **West Bengal**, where approximately **2.3M is through Bank** and only **0.2M through Post Office**.

At the scheme level, **IGNOAPS** is the dominant scheme in payment activity, with approximately **7.8M Bank, 2.5M Cash, and 1.4M Post Office payments**. This suggests that the payment channel mix varies considerably by scheme.

Key Diagnostic Insight

The major issue is not simply the number of beneficiaries. The dashboard reveals a difference in how pension payments are delivered across states and schemes. This suggests that **regional infrastructure, beneficiary preferences, accessibility, and scheme-specific delivery mechanisms** may influence payment methods.

3. Predictive Analysis — What May Happen?

- Given that **90.18% of beneficiaries are rural**, future pension delivery demand is likely to remain strongly concentrated in rural areas. Bank-based payments may continue to grow if digital and banking infrastructure becomes more accessible.
- States with strong reliance on cash, particularly **Odisha with around 4.0M Cash payments**, could potentially see a shift toward bank-based or other digital payment channels if financial access and beneficiary adoption improve.
- The **21M Unpaid Beneficiaries KPI** is also an important monitoring indicator. If this measure remains high over time, it may indicate a need for closer monitoring of pension delivery and beneficiary payment status.

Future analysis should track:

- Year-wise growth in beneficiaries
- Year-wise pension disbursement
- Changes in unpaid beneficiary counts
- Movement from Cash to Bank payments
- Rural vs Urban beneficiary trends

4. Prescriptive Analysis — What Should Be Done?

Based on the dashboard findings, several development actions can be recommended.

1. Strengthen Rural Pension Delivery

Since **39M beneficiaries (90.18%)** are rural, government agencies should prioritize **rural banking access, local service centers, digital payment support, and beneficiary awareness programs.**

2. Improve Payment Method Accessibility

Bank payments account for approximately **14M**, making them the leading payment method. However, some states show substantial reliance on Cash. Payment strategies should be designed according to **local infrastructure and beneficiary accessibility** rather than applying a single approach everywhere.

3. Focus on High-Cash States

Odisha is a major area for further investigation, with approximately **4.0M Cash payments** versus **1.1M Bank payments**. A targeted study could identify whether beneficiaries face barriers to banking access and whether optional digital/bank-based payment channels can be expanded.

4. Monitor Unpaid Beneficiaries

The dashboard reports **21M Unpaid Beneficiaries**. This KPI should be monitored regularly at **state, district, and scheme levels** to identify where payment delivery requires attention.

5. Prioritize Major Schemes

IGNOAPS, IGNWPS, and IGNDPS are among the leading schemes by beneficiary records, with **1,175, 1,079, and 1,069** respectively. These schemes should receive strong monitoring because improvements in their delivery can have a significant impact on overall pension coverage.

Overall Key Insights

- The dashboard covers **43M beneficiaries** across **30 states, 625 districts, and 33 pension schemes**, providing broad coverage of the pension system.
- Beneficiary coverage is highly concentrated in **rural areas (90.18%)**, with urban beneficiaries accounting for **9.82%**, highlighting the importance of rural pension delivery.
- **Bank payments (14M)** are the leading disbursement channel, followed by **Cash (6M), Post Office (2M), and Money Order (62K)**.

- The dashboard shows **22M pension disbursements** and **21M beneficiaries shown as unpaid**, indicating a significant gap that requires further monitoring and validation.
- Payment patterns vary considerably across **states and pension schemes**. For example, Odisha shows high reliance on **Cash payments (around 4.0M)** compared with **Bank payments (around 1.1M)**.
- The major schemes **IGNOAPS, IGNWPS, and IGNDPS** have the highest beneficiary record counts, indicating their importance in the overall pension ecosystem.

Overall Recommendations

Strengthen Rural Pension Services: Prioritize banking infrastructure, local service centers, and payment accessibility because **90.18% of beneficiaries are from rural areas**.

Monitor Unpaid Beneficiaries: Regularly track the **21M unpaid beneficiaries KPI** at state, district, and scheme levels to identify and address payment gaps.

Optimize Payment Channels: Encourage efficient and accessible **bank-based payment systems** while ensuring alternatives remain available for beneficiaries with limited banking access.

Investigate State-Level Differences: Conduct deeper analysis of states with high cash dependency, such as **Odisha**, to understand infrastructure and accessibility challenges.

Focus on Major Schemes: Closely monitor **IGNOAPS, IGNWPS, and IGNDPS** because they represent a significant share of beneficiary records.

Track Trends Over Time: Continuously monitor changes in beneficiary numbers, pension disbursement, unpaid beneficiaries, and payment methods to evaluate program performance and improve pension delivery.

Final Project Conclusion

The analysis highlights a predominantly rural pension beneficiary base, significant variation in payment methods across states and schemes, and a substantial number of unpaid beneficiaries. The findings suggest that strengthening rural payment infrastructure, improving accessibility to efficient payment channels, and closely monitoring unpaid beneficiaries can help enhance the effectiveness and reach of pension delivery.